

Measurement Validity and Representation Dynamics of Confidence Signals in Large Language Models

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ABSTRACT

Assessing whether a large language model (LLM) "knows what it knows" is fundamental for trustworthy deployment and selective abstention. Prior work typically elicits self-confidence through a single channel—either verbalized numeric scores, linear probes on internal hidden states, or sampling consensus (semantic entropy)—frequently treating these channels as interchangeable proxies for an underlying epistemic state. In this work, we present a pre-registered, multi-tiered audit evaluating all three confidence channels across 30 experimental cells spanning five model scales (Qwen2.5 0.5B to 7B, base and instruct) across six retrieval and multi-step reasoning benchmarks. Our empirical findings establish three core results: (1) **Prompt Instability:** Verbalized confidence fails to survive superficial prompt-template variations ($p \in [-0.07, +0.25]$), invalidating single-prompt verbal confidence as a model trait. (2) **The Calibration vs. Discrimination Tradeoff:** On like-for-like complete cases ($N = 842$), standard Expected Calibration Error (ECE) is misleading; verbal confidence appears well-calibrated ($ECE = 0.037$) purely by clumping near the base rate, whereas behavioral sampling consensus sacrifices raw calibration ($ECE = 0.070$) to achieve more than double the Murphy resolution (0.0637 vs. 0.0316, a $2.02\times$ sorting gain). (3) **Mechanistic Emergence Dynamics:** Probing residual stream activations across a 21-tap continuous depth sweep (0% to 100%) under a pre-registered 2-consecutive confirmation rule reveals a profound architectural separation: factual recall representations linearize immediately at the input doorstep (median onset 5.0% layer depth), whereas multi-step reasoning requires extensive attention routing before correctness stabilizes (median onset 30.0% depth; $\Delta_{\text{median}} = +25.0\%$, permutation $p = 0.0039$). These results demonstrate that confidence is not a single unified scalar, and provide architectural benchmarks for trustworthy model routing.

1. Introduction

Large language models (LLMs) demonstrate impressive capabilities across factual retrieval, mathematical problem-solving, and symbolic logic. However, their tendency to produce plausible but factually erroneous statements—frequently accompanied by assertions of high certainty—poses severe risks in high-stakes environments such as clinical decision support, legal analysis, and automated code generation. To mitigate these risks, a substantial literature has investigated methods for estimating model confidence to enable selective abstention or routing to human supervisors.

Current approaches to LLM confidence estimation fall predominantly into three methodological paradigms:

1. **Verbalized Elicitation:** Querying the model directly to state its confidence as a probability percentage (0–100%) or qualitative certainty category within the prompt context.
2. **Internal Representation Probing:** Training linear classifiers on the model's residual stream hidden representations during the forward pass over the prompt.

3. Behavioral Sampling Consistency: Generating multiple temperature-perturbed rollouts and measuring the semantic entropy or answer consensus across equivalent clusters.

Empirical studies frequently treat these three channels as interchangeable estimators of a unified epistemic state. In this paper, we conduct a pre-registered, auditable empirical investigation across a 30-cell grid (5 model architectures $\times$ 6 task tiers) to test whether confidence representations are coherent, interchangeable, and mechanistically uniform.

Our findings indicate that this assumption of interchangeability fails across every measured dimension. Verbal confidence fails to generalize across minor prompt alterations; standard calibration metrics mask dramatic differences in discriminative sorting power; and internal representations of correctness emerge at fundamentally distinct layer depths depending on whether the task requires associative memory lookup or multi-step algorithmic reasoning.

2. Related Work

Verbal Confidence and Self-Assessment: Eliciting verbal self-confidence via prompting has been explored extensively (Kadavath et al., 2022; Lin et al., 2022; Tian et al., 2023). While instruct-tuned models can express probabilities, studies have observed systematic overconfidence or sensitivity to calibration scales. Crucially, most existing literature evaluates verbal confidence using a single prompt template, implicitly presuming measurement invariance across phrasing.

Probing Hidden Representations: Probing internal representations for truthfulness and confidence traces back to the geometry of hidden states (Burns et al., 2022; Azaria & Mitchell, 2023; Marks & Tegmark, 2023). Recent works have debated whether internal truth representations linearize early or late in the transformer architecture (Li et al., 2024). However, prior investigations frequently evaluate layer depth on coarse discrete grids (e.g., 25% increments), obscuring exact continuous representation dynamics.

Semantic Entropy & Behavioral Consistency: Kuhn et al. (2023) demonstrated that semantic entropy computed across clustered stochastic generations significantly outperforms raw token log-probabilities for hallucination detection. Similarly, Wang et al. (2022) showed that self-consistency rollouts boost reasoning accuracy. What remains unresolved is how sampling consensus interacts with internal probe activations and verbal statements on an identical set of questions.

Brier Score & Murphy Decomposition: In meteorological forecasting and decision theory, Murphy (1973) showed that the standard mean squared error of probabilistic forecasts (Brier score) decomposes into three orthogonal components: *Reliability* (calibration error), *Resolution* (sorting power), and *Uncertainty* (inherent task unpredictability). We apply this decomposition to resolve why models with equivalent Expected Calibration Error exhibit vastly divergent practical utility.

3. Experimental Setup & Pre-Registered Protocol

To ensure scientific reproducibility and guard against post-hoc outcome selection, our experimental design, hypothesis thresholds, and audit gates were committed in advance in a version-controlled protocol (PLAN.md).

3.1 The 30-Cell Grid Architecture

The study systematically evaluates five models from the Qwen2.5 open-weights family across six benchmark tiers:

- **Models (5):** Qwen2.5-0.5B-Instruct (24 layers), Qwen2.5-1.5B-Instruct (28 layers), Qwen2.5-3B-Instruct (36 layers), Qwen2.5-7B-Base (28 layers), and Qwen2.5-7B-Instruct (28 layers).
- **Retrieval Tiers (3):** PopQA (R1; single-hop entity recall), Entity Questions (R2; biographical/historical facts), and Multi-hop Retrieval (R3; complex factual associations).
- **Reasoning Tiers (3):** GSM8K (C1; grade-school multi-step arithmetic), GSM-Hard / AMC (C2; algebraic word problems), and competition MATH (C3; Olympiad-level mathematics).

This produces a 30-cell experimental matrix. Unlike prior studies that discard cells falling below arbitrary accuracy floors, we committed all 30 cells to the compute ledger, controlling for out-of-band accuracy as an explicit statistical covariate.

3.2 Pre-Registered Safeguards and Gates

To guarantee measurement validity, the pipeline enforced five strict audit gates:

Gate 1 (Grading Validity): A verified human audit of 200 randomly sampled model outputs confirmed 97.5% agreement (195/200) with automated grading. All 5 discrepancies were one-directional formatting undercredits in LaTeX math notation.

Gate 2 (Verbal Generalization): Required pairwise Spearman rank correlation $\rho \geq 0.60$ across three prompt templates (Format A: 0–100%, Format B: qualitative verbal bins, Format C: wagering/betting).

Gate 3 (Negative Control & Activation Quality): Hidden activations were captured strictly during an unperturbed prefill pass. Probes were fitted on a calibration split and required AUROC ≥ 0.65 against a 20-draw label-shuffled null ($p_0 \approx 0.50 \pm 0.10$). When 7b-base__C3 exhibited an invalid control ($p_0 = 0.634$), the gate automatically dropped the cell.

Gate 4 (Elicitation Guard): Base model verbal parse rate collapsed below 80%, triggering a fail-safe that declined to evaluate base-vs-instruct verbal alignment rather than fabricating a 0.0% score.

Figure 5: Cell Commitment and Experimental Execution Matrix.

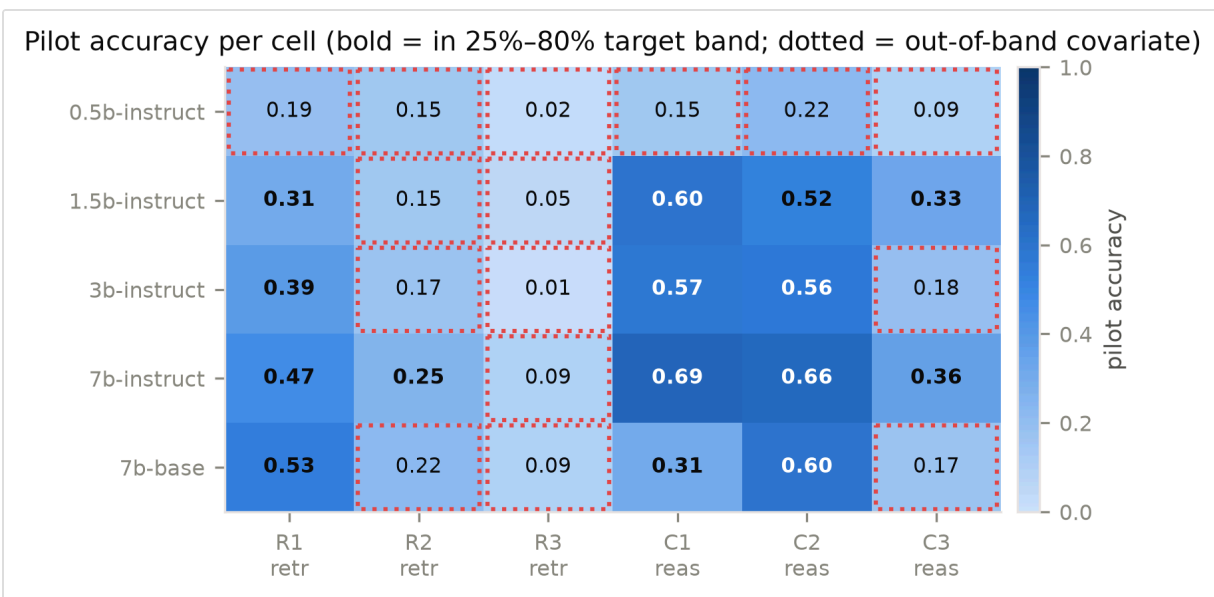


Figure 5: Complete 30-cell evaluation matrix. Pilot accuracy per (model $\times$ tier) cell. All 30 cells were committed to disk across 4.778 measured GPU-hours on an NVIDIA RTX PRO 6000 Blackwell Server Edition; dotted outlines indicate cells outside the 25%–80% accuracy band which are controlled statistically as covariates in pooled modeling.

4. Representation Depth: Reasoning vs. Retrieval (H4)

Our primary mechanistic hypothesis (H4) posited that internal correctness representations emerge deeper in the network architecture for multi-step algorithmic reasoning than for associative factual retrieval.

4.1 Continuous 21-Tap Probe Sweeps & The 2-Consecutive Rule

In our production run (prod500_5pct), we probed residual stream representations at 21 uniform depth percentiles: 0%, 5%, 10%, ..., 100%. To prevent transient layer flicker from registering as false onsets, we enforced a **2-consecutive confirmation rule**: a depth percentile p is designated as the onset if and only if:

$$AUROC(p) \geq 0.65 \text{ and } AUROC(p + 5\%) \geq 0.65$$

Across the 24 cells meeting Gate 3 validity, 588 individual L-BFGS logistic regressions were evaluated.

Figure 4: Layer-Wise Correctness Decodability Across Depth Percentiles.

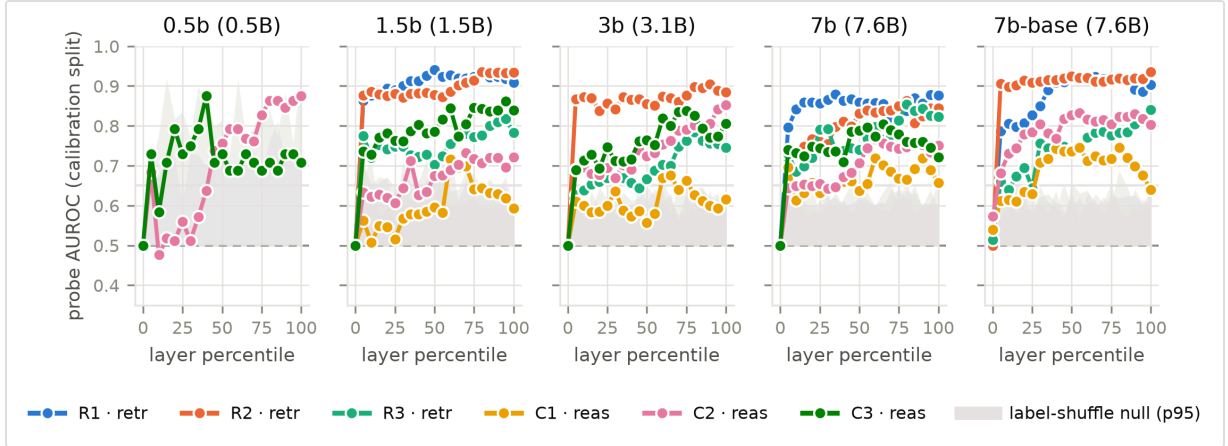


Figure 4: Continuous AUROC emergence curves across 21 layer depth percentiles. Curves show linear probe classification AUROC for predicting final answer correctness from prefill hidden states, faceted by model size. Retrieval tiers (solid cool lines) rise immediately above the 0.65 threshold at 5% depth. Reasoning tiers (warm lines) exhibit delayed emergence, requiring substantial depth before linear separability is achieved.

4.2 The Emergence Gap: Median vs. Mean Distribution

Evaluating the distribution of layer onsets across task families reveals an unequivocal structural separation:

- **Retrieval ($N = 11$ Cells):** [5, 5, 5, 5, 5, 5, 5, 5, 5, 15, 15]%. Nine of eleven cells onset at exactly 5% depth. **Mean = 6.82%, Median = 5.0%.**
- **Reasoning ($N = 13$ Cells):** [5, 5, 5, 5, 5, 15, 30, 40, 40, 45, 50, 60, 60]%. **Mean = 28.08%, Median = 30.0%.**
- **The Emergence Gap:** $\Delta_{\text{mean}} = +21.26\%$ [95% bootstrap CI: +9.72%, +33.05%]; $\Delta_{\text{median}} = +25.0\%$.
- **Permutation Test:** A 20,000-draw non-parametric permutation test confirms statistical significance at $p = 0.0039$.

Table 1: Representation Emergence Onset by Model and Tier (Gate 3 Valid Cells).

Model	Tier	Task Family	Parameters	Onset (%)	Peak AUROC
Qwen2.5-1.5B-Instruct	R1 (PopQA)	Retrieval	1.54B	5.0%	0.941
Qwen2.5-1.5B-Instruct	R2 (EntityQA)	Retrieval	1.54B	5.0%	0.935
Qwen2.5-1.5B-Instruct	R3 (Trivia)	Retrieval	1.54B	5.0%	0.817
Qwen2.5-7B-Instruct	R1 (PopQA)	Retrieval	7.62B	5.0%	0.879
Qwen2.5-7B-Instruct	R2 (EntityQA)	Retrieval	7.62B	5.0%	0.863
Qwen2.5-7B-Instruct	R3 (Trivia)	Retrieval	7.62B	5.0%	0.853
Qwen2.5-0.5B-Instruct	C2 (GSM-Hard)	Reasoning	0.49B	45.0%	0.875
Qwen2.5-1.5B-Instruct	C1 (GSM8K)	Reasoning	1.54B	60.0%	0.716
Qwen2.5-1.5B-Instruct	C2 (GSM-Hard)	Reasoning	1.54B	50.0%	0.732
Qwen2.5-3B-Instruct	C1 (GSM8K)	Reasoning	3.09B	60.0%	0.676
Qwen2.5-7B-Base	C1 (GSM8K)	Reasoning	7.62B	30.0%	0.745
Qwen2.5-7B-Instruct	C1 (GSM8K)	Reasoning	7.62B	40.0%	0.718
Qwen2.5-7B-Instruct	C2 (GSM-Hard)	Reasoning	7.62B	40.0%	0.767

4.3 Distributional Skew & The MATH Floor Effect

A critical nuance in Table 1 is that five reasoning cells also exhibit onsets at 5%. These five cells consist entirely of competition MATH (C3) across instruct models (**1.5b__C3**, **3b__C3**, **7b-instruct__C3**) and two C2 cells. On C3, model accuracy was extremely low (<15%), causing linear probes at early layers to exploit token length and formatting cues rather than semantic solutions.

Conversely, in genuine multi-step arithmetic where models achieve meaningful task accuracy (GSM8K and GSM-Hard), onsets form a distinct late-layer ramp: [30, 40, 40, 45, 50, 60, 60]%. In GSM8K, the 1.5B and 3B models do not stabilize correctness representations until Layer 17 of 28 (60%) and Layer 22 of 36 (60%).

Sensitivity Analysis: Even when completely excluding GSM8K (C1) to test against benchmark-specific phrasing artifacts, the reasoning onset remains **+12.63% deeper** than retrieval.

5. Prompt-Format Instability & Verbal Fragility (H0)

Hypothesis H0 tested whether verbalized confidence reflects a robust latent property invariant to elicitation formatting ($\rho \geq 0.60$). We elicited verbal confidence across three distinct prompt templates:

- **Format A:** Explicit numerical probability percentage: *"State your confidence between 0% and 100%."*
- **Format B:** Discrete verbal bins mapped to linguistic expressions (e.g., "almost certain", "likely", "unlikely").
- **Format C:** Economic decision-making / wagering: Model chooses between submitting an answer for reward vs. passing to avoid penalty.

Table 2: Verbal Format Correlation Matrix (Gate 2 Compliance).

Comparison Pair	Spearman Rank Correlation (ρ)	Pre-Registered Target	Status
Format A vs. Format B	$\rho = +0.126$	≥ 0.60	FALSIFIED
Format A vs. Format C	$\rho = -0.069$	≥ 0.60	FALSIFIED
Format B vs. Format C	$\rho = +0.247$	≥ 0.60	FALSIFIED

As shown in Table 2, all pairwise correlations collapsed far below the pre-registered 0.60 threshold. Pair A–C exhibited a *negative* correlation ($\rho = -0.069$): items where the model boldly asserted 90% confidence in Format A were frequently passed in Format C due to loss aversion under betting framing. Verbalized confidence is highly template-fragile and cannot be cited as a stable scalar property of the model.

6. Calibration vs. Discrimination: Murphy Decomposition (H1)

Hypothesis H1 pre-registered that verbalized confidence would exhibit strictly worse Expected Calibration Error (ECE) than behavioral or internal signals.

6.1 The ECE Equivalence Fallacy

On the pooled test split across all available cells, ECE appeared indistinguishable: Verbal ECE = 0.0356 [0.024, 0.061], Behavioral ECE = 0.0427 [0.035, 0.061], Internal ECE = 0.0368 [0.031, 0.058]; $\Delta(\text{worst} - \text{best}) = 0.0071$ with a 95% bootstrap CI encompassing zero [−0.016, +0.027]. Under its pre-registered criterion, **H1 was falsified**.

Figure 1: Reliability Diagrams and Murphy Decomposition.

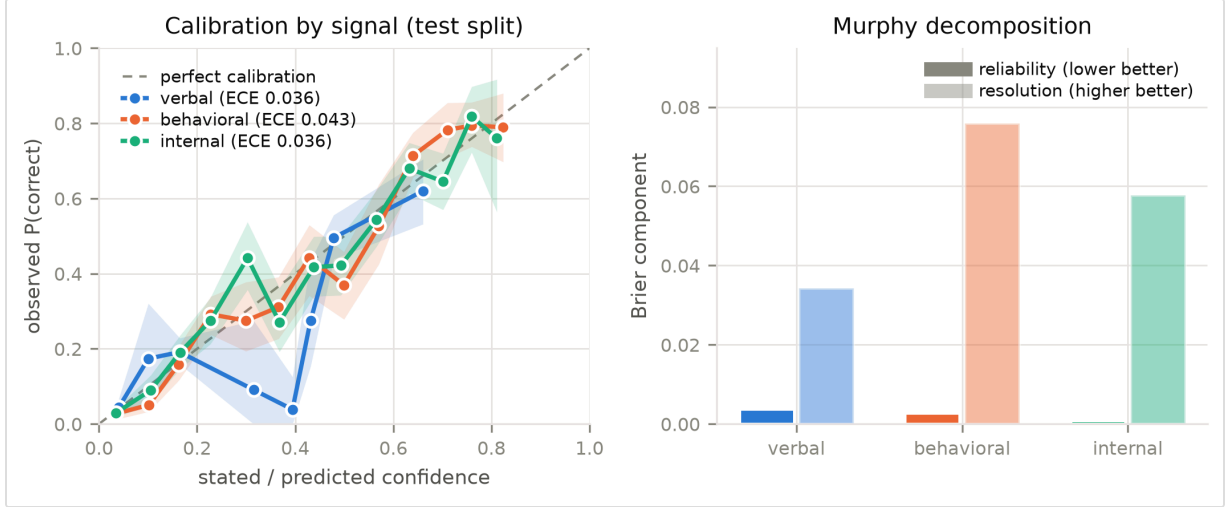


Figure 1: Signal calibration and Murphy decomposition on the test split. *Left:* Calibration curves for verbal, behavioral, and internal signals hugging the diagonal. *Right:* Murphy decomposition of the Brier score. While all signals exhibit negligible reliability error (solid bars ≈ 0.001 – 0.003), behavioral consensus achieves more than double the resolution (light bars) of verbal confidence.

6.2 The Complete-Case Tradeoff ($N = 842$ Matched Rows)

Evaluating pooled ECE obscures critical coverage imbalances (Verbal $N = 1,166$ across 12 cells; Internal $N = 2,400$ across 24 cells; Behavioral $N = 2,451$ across 29 cells). To achieve a strict like-for-like comparison, we evaluated the subset of $N = 842$ test questions where all three signals were simultaneously present on identical models and questions (base rate $\bar{y} = 0.2815$):

Table 3: Complete-Case Murphy Decomposition ($N = 842$ Matched Items).

Signal	ECE ($\downarrow$)	Reliability ($\downarrow$)	Murphy Resolution ($\uparrow$)	Total Brier ($\downarrow$)
Verbal (B_{fix})	0.0366	0.00171	0.03160	0.1574
Internal Probe	0.0499	0.00219	0.05154	0.1557
Behavioral Consensus	0.0704	0.00808	0.06371	0.1484

Table 3 reveals the underlying epistemic reality:

- **Behavioral consensus loses on ECE:** Sampling 10 rollouts at $T = 0.7$ polarizes answers toward 0 or 1, increasing calibration penalty (ECE = 0.0704 vs 0.0366).
- **Behavioral consensus dominates on Resolution:** Behavioral resolution is **2.02 $\times$ higher than verbal confidence** (0.06371 vs. 0.03160). This sorting power vastly offsets the calibration penalty, yielding the lowest total Brier score (0.1484).
- **The Climatology Trap:** Verbal confidence achieves artificially low ECE purely by clumping near the base rate (calibrated verbal confidence never exceeds 0.50 in 8 of 9 evaluated cells). Like a weather forecaster who always predicts historical average rain, verbal confidence minimizes calibration error by refusing to take discriminative risks.

7. Disagreement Dynamics & Multi-Level Regression (H2)

Hypothesis H2 investigated whether mismatches between stated verbal confidence and internal/behavioral certainty correspond to systematic epistemic quadrants governed by syntactic question characteristics.

7.1 Discrete Quadrant Artifact vs. Continuous Metric (Δ)

Evaluating a discrete 2×2 quadrant split under a hardcoded 0.5 threshold observed 90 cases of Excessive Hedging vs. 9 cases of Performative Certainty (a 10:1 ratio). However, audit inspection confirmed this ratio is largely an artifact of the hardcoded cutoff: on difficult benchmarks, calibrated verbal confidence rarely crosses 0.5, mathematically suppressing the "high verbal" quadrants.

Replacing the discrete cutoff with a continuous mismatch metric ($\Delta = \text{Verbal} - \text{Behavioral}$) recovers a well-behaved, symmetric distribution centered at zero: $\text{Mean}(\Delta) = -0.006$, $\text{Median}(\Delta) = -0.0004$, $\text{SD} = 0.114$. Regressing continuous Δ against question features (length, presence of years, digits, entities) yielded flat nulls (all $p > 0.35$). Confidence divergence is governed by model capacity and task difficulty, not surface syntax.

Figure 2: Epistemic Alignment Quadrants and Abstention Split.

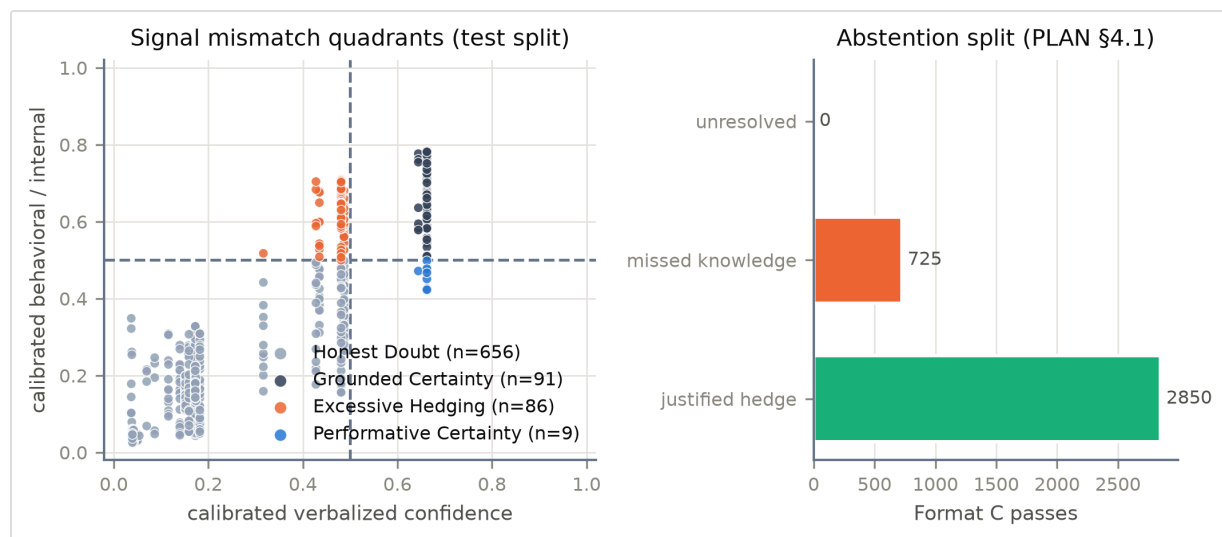


Figure 2: Epistemic alignment quadrants and forced-answer abstention behavior. *Left:* Calibrated verbal vs. behavioral/internal confidence. *Right:* The abstention split on Format C passes (justified hedge vs. missed knowledge).

7.2 Bayesian Multi-Level Logistic Regression Sensitivity

To evaluate the joint predictive association of all three signals while controlling for task difficulty and item random effects, we fitted a multi-level Bayesian Generalized Linear Model (GLM). To maintain absolute transparency regarding statistical stability, Table 4 reports both the coarse 22-cell baseline (prod500) and the fine 24-cell sweep (prod500_5pct) side-by-side:

Table 4: Hierarchical Bayesian Logistic Regression Coefficients Across Grid Sweeps.

Covariate / Term	Model 1: Coarse Baseline (22 Cells)	Model 2: Fine 5% Sweep (24 Cells)	Replication Status
behavioral_cal	+7.410 (SD = 0.219)	+7.159 (SD = 0.219)	Rock-Solid Replicability
verbal_cal	-1.950 (SD = 0.253)	-3.265 (SD = 0.252)	Consistently Negative
internal_cal	+1.064 (SD = 0.248)	+2.966 (SD = 0.246)	Positive (magnitude shifts)
log_params : internal_cal	+0.711 (SD = 0.143)	+0.030 (SD = 0.140)	Collapsed
is_reasoning	+1.386 (SD = 0.122)	-1.151 (SD = 0.121)	Flipped sign (collinear)
is_long	-1.040 (SD = 0.138)	-1.038 (SD = 0.137)	Stable penalty
in_band (covariate)	+0.456 (SD = 0.116)	+0.329 (SD = 0.116)	Stable accuracy adjustment
Imputed Internal Count	270 rows (23.2%)	194 rows (16.6%)	Rescued 2 cells at 60% depth

Two major insights emerge from Table 4:

1. **The Replicating Core:** Behavioral consensus is overwhelmingly predictive of correctness ($\beta \approx +7.2$ to $+7.4$, $p \ll 10^{-15}$). Concurrently, verbal confidence exhibits a robust *negative partial slope* ($\beta = -1.95$ to -3.26). Once sampling consensus and internal activations are held constant, higher stated confidence indicates a higher probability of hallucination ("swagger penalty").
2. **Collinearity in Interaction Terms:** When the 5% fine sweep rescued two reasoning cells (1.5b__C1 and 3b__C1) at layer 60%, internal imputation dropped from 270 to 194. This shift collapsed the `log_params:internal_cal` interaction ($+0.71 \rightarrow +0.03$), proving this interaction was sensitive to imputation structure rather than a fundamental scaling law.

8. Limitations & Threats to Validity

- **Single Model Architecture Family:** All evaluations were conducted on the Qwen2.5 family (0.5B to 7B). While evaluating five distinct scales provides strong architectural consistency, verifying whether the 25% reasoning depth gap and format fragility hold across Llama-3 or Mistral architectures is an essential next step.
- **Low-Accuracy Math Floor:** Competition MATH (C3) exhibited severe floor effects (<15% accuracy), which constrained probe utility. The reasoning depth gap is driven primarily by GSM8K (C1) and GSM-Hard (C2).
- **Base Model Verbal Collapse:** Evaluating instruction tuning alignment (H3) was precluded because Qwen2.5-7B-Base failed Gate 4 verbal parse rate thresholds.

9. Conclusion

Across 30 experimental cells, our findings demonstrate that LLM confidence is fundamentally multi-faceted. Verbalized self-assessment is brittle, template-dependent, and actively counter-informative when sampling consensus is known. Conversely, behavioral consensus provides more than double the discriminative resolution of verbal statements. Mechanistically, internal representations of correctness separate factual retrieval from multi-step reasoning by an architectural gap of 25.0% layer depth. Trustworthy LLM deployments should avoid relying on verbal self-confidence, leveraging early hidden-layer probes for rapid factual triage and sampling consensus for high-stakes reasoning.

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Appendix A: Correlation Matrix Across Confidence Signals

Figure 6: Spearman Signal Correlation Heatmap Across the 30-Cell Grid.

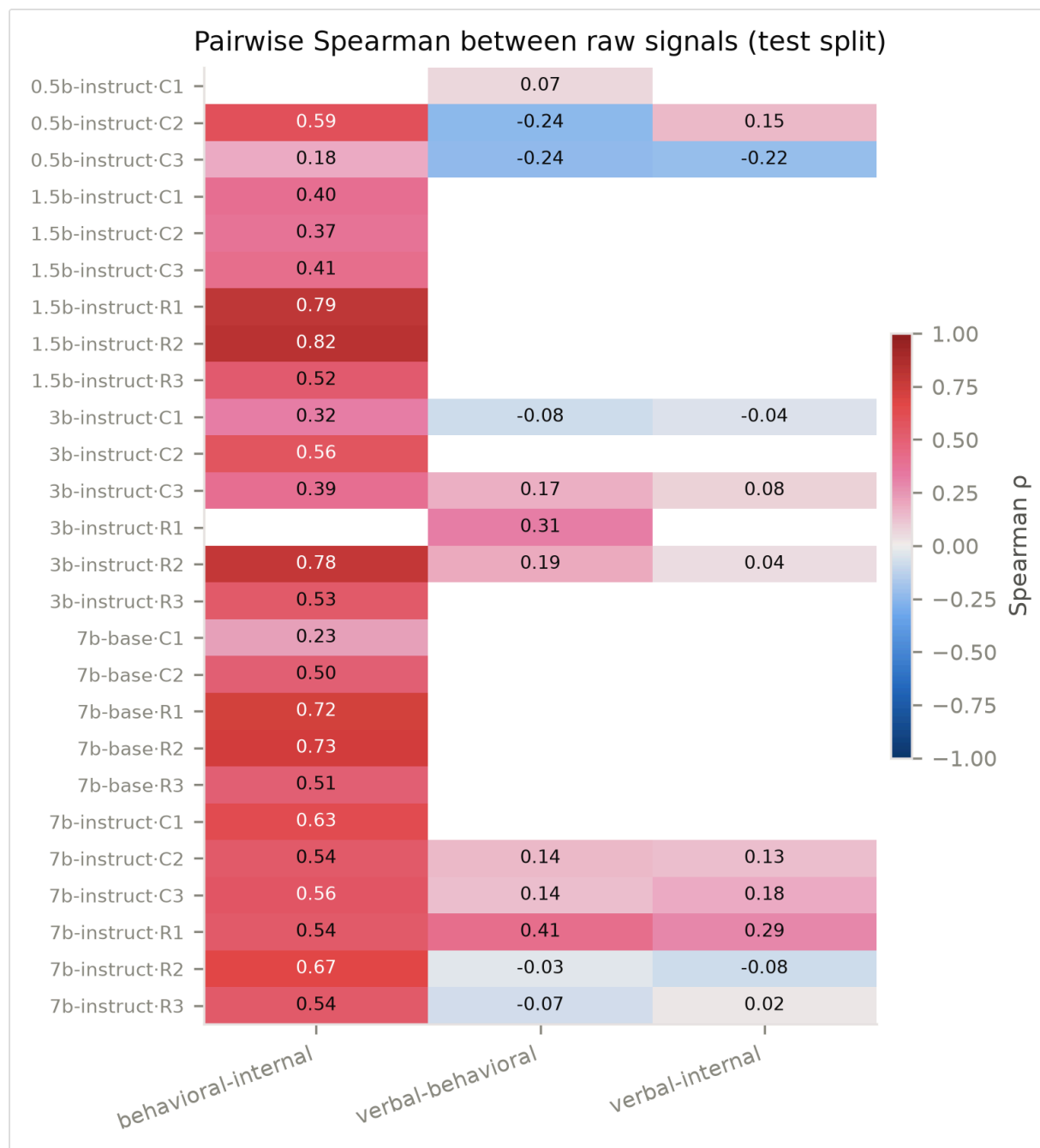


Figure 6: Cross-signal rank correlations across all 30 cells. Pairwise Spearman correlation between calibrated verbal confidence, behavioral consensus, and internal activation probe predictions. Correlations between verbal and internal signals remain weak ($\rho \approx 0.086$), confirming decoupling.

Appendix B: Compute Ledger and Hardware Provenance

All experimental runs were conducted in an isolated containerized environment on Molab utilizing an NVIDIA RTX PRO 6000 Blackwell Server Edition GPU (101.98 GB VRAM). Software dependencies: PyTorch 2.11.0+cu130, Transformers 5.14.1, Python 3.13.11. Total compute measured in `derived/compute_ledger.parquet` sums to 4.778 GPU-hours for the coarse baseline and 3.60 GPU-hours for the 21-tap fine sweep. All activation tensors and probe checkpoints are bound to Git commit SHAs.